

Model Question

Section A (Answer any TWO questions) $[2 \times 10 = 20]$

1. Classify the operators available in C (e.g., arithmetic, relational, etc.). Explain any four classes with suitable examples. [10]
2. State key properties of arrays. Write a program that inputs ages for 500 people and then displays (i) the average age and (ii) how many ages fall within 25 to 30 (inclusive). [10]
3. Differentiate between standard library functions and user-defined functions. Then write a C program that swaps two variables using the call-by-reference idea. [10]

Section B (Answer any EIGHT questions) $[8 \times 5 = 40]$

4. Describe the typical layout/parts of a C program (headers, main, declarations, etc.). [5]
5. Explain formatted input/output functions and why they are used. Include examples. [5]
6. Write a program to print the first 50 prime numbers. [5]
7. Generate the series: $(2 \times 3), (3 \times 5), (4 \times 7), (5 \times 9), \dots$ up to 25 terms, but skip printing the 7th term. [5]
8. Demonstrate recursion with a suitable example program and explanation. [5]
9. Create a structure **STUDENT** with fields like ID, name, address, and CGPA. Initialize data for 100 students and display those whose address is “KTM” and CGPA lies between 3.5 and 4.0. [5]
10. Explain common file opening modes (read, write, append, etc.) and their use-cases. [5]
11. Using graphics functions, write a program that draws any two simple shapes of your choice. [5]
12. Write short notes on: (i) global variables and (ii) debugging. [5]

Model Question

Section A (Answer any TWO questions) $[2 \times 10 = 20]$

1. Define a structure and explain what a nested structure means. Then write a program that takes n and checks whether the n th Fibonacci number is prime. Use separate user-defined functions for Fibonacci generation and primality testing. [10]
2. Explain how arrays and pointers relate in C. Also distinguish call-by-value vs. call-by-reference using a suitable example program. [10]
3. Differentiate source code and object code. Create a structure `Book` containing fields such as name, price, and author. Input 10 records and print authors of books with price above 1000. [10]

Section B (Answer any EIGHT questions) $[8 \times 5 = 40]$

4. Describe file-handling I/O functions in C (formatted/unformatted where relevant) and show basic syntax. [5]
5. Read a $P \times Q$ integer matrix and display the largest element from each row. [5]
6. Compute factorial of a number using recursion. [5]
7. Check whether an input word/string is a palindrome. [5]
8. List operator categories in C and explain any three with examples. [5]
9. Predict the output for this logic: initialize `i=0`; for `k` from 5 down to 0, do `i=i+k`; then print `i`. [5]
10. Using a function, compute the sum of the first 10 even numbers. [5]
11. Explain dynamic memory allocation and illustrate with a short program (e.g., using `malloc` / `free`). [5]
12. Initialize an array of size 10 and sort it in ascending order. [5]

Model Question

Section A (Answer any TWO questions) $[2 \times 10 = 20]$

1. Explain the difference between `exit(0)` and `exit(1)`. Discuss why nested structures are useful (with an example). Then write a program to compute x^y without using the built-in power function. [10]
2. Explain why `break` and `continue` are needed. Define actual vs. formal parameters with examples. Finally, given a small C snippet with issues (missing semicolons, incorrect `scanf` format/quotes, mismatched names inside a function, returning a value from a `void` function, etc.), identify the errors. [10]
3. Write a menu-driven program where a user selects options using a character choice to: (A) check odd/even, (B) check positive/negative, (C) compute factorial, or (D) exit. Continue showing the menu until the exit choice is given. [10]

Section B (Answer any EIGHT questions) $[8 \times 5 = 40]$

4. Swap two integers without using a third temporary variable. Justify with an example. [5]
5. Find the sum of digits of an integer using recursion. [5]
6. Distinguish constants and literals. Why must we specify data types in programming? [5]
7. Given an array, write a program to find the second-largest element. [5]
8. Create a structure `Employee` (name, address, salary, age). Display names of employees aged 40–50 who live in Kathmandu. [5]
9. State one advantage and one limitation of pointers. Explain how pointers can be passed to functions as arguments. [5]
10. Read integers from `Num.txt`, extract only prime numbers, and write those primes into `Prime.txt`. [5]
11. Explain why a union can be preferable to a structure in some cases. Also list four common string library functions with their prototypes. [5]
12. Write short notes on: (i) local, global, and static variables; (ii) the conditional (ternary) operator. [5]

Model Question

Section A (Answer any TWO questions) [2×10 = 20]

1. What are jump statements in C? Explain each (e.g., **break**, **continue**, **goto**, **return**) with examples. Then write a program to test whether a number is prime. [10]
2. Explain any three common string functions. Then write a program to check whether two matrices are identical. [10]
3. Define structure and illustrate nested structure. Create a **book** structure containing fields like name, author, publisher. Read data for 50 books and display names of those books published by a specified publisher (e.g., “XYZ”). [10]

Section B (Answer any EIGHT questions) [8×5 = 40]

4. Explain the structure of a C program with a simple example. [5]
5. What is a variable? How is it different from a constant? How do you write comments in C? [5]
6. Explain formatted I/O functions with examples. [5]
7. Using your own function, write a program to find the sum of two numbers. [5]
8. Write a program to find the largest among three input numbers. [5]
9. Explain dynamic memory allocation with an example. [5]
10. Write a simple program that reads data from a file and displays it. [5]
11. Using graphics functions, draw a line. [5]
12. Write short notes on: (i) brief history of C, (ii) bitwise operators. [5]

Model Question

Section A (Answer any TWO questions) $[2 \times 10 = 20]$

1. Explain looping. Describe the `while` loop with example and compare it with `do-while`. Then write a program to compute sum and average of the first n natural numbers. [10]
2. State benefits of arrays and compare 1D vs 2D arrays. Then write a program to compute transpose of a matrix. [10]
3. Explain structure and compare it with union. Create a structure `course` (name, code, credit_hour). Read 5 courses and display those with credit hours greater than 3. [10]

Section B (Answer any EIGHT questions) $[8 \times 5 = 40]$

4. Explain flowcharts with an example and mention their advantages. [5]
5. What is a data type and why is it required? Explain any three basic data types with examples. [5]
6. Explain unformatted I/O in C and where it is used. [5]
7. Write a program to display the first n prime numbers. [5]
8. Using a user-defined function, compute the product of two integers. [5]
9. Define pointers. Explain (conceptually) how a function can return a pointer, with a simple example. [5]
10. Explain common file I/O functions with examples. [5]
11. Draw a circle using graphics functions. [5]
12. Write short notes on: (i) compilation vs execution, (ii) operator precedence and associativity. [5]

Model Question

Section A (Answer any TWO questions) $[2 \times 10 = 20]$

1. Define looping. Discuss different looping constructs in C with suitable examples. [10]
2. Define arrays and list benefits. Write a program to add two matrices using arrays. [10]
3. Why do we use data files? Explain file opening modes. Write a program that reads from `input.txt` and writes the content into `output.txt`. [10]

Section B (Answer any EIGHT questions) $[8 \times 5 = 40]$

4. Explain logical operations/operators in C with examples. [5]
5. Explain `break` with example and differentiate it from `continue`. [5]
6. Write a program to check whether an integer is even or odd. [5]
7. Write a program to compute the sum of the first 10 odd numbers. [5]
8. What are preprocessor directives? Explain the `#define` directive with an example. [5]
9. Discuss any five string library functions and their use. [5]
10. Explain dynamic memory allocation and illustrate how `malloc()` is used. [5]
11. Define structures and create a `rectangle` structure with length and breadth. [5]
12. Write short notes on: (i) benefits of data files, (ii) graphics functions. [5]

Model Question

Section A (Answer any TWO questions) $[2 \times 10 = 20]$

1. Explain the standard structure of a C program with an example. [10]
2. Discuss different forms of `if` statements with examples. Also compare `if-else` with `switch`. [10]
3. Explain structures and compare them with arrays. Create a `student` structure (name, roll, percentage) and display students scoring at least 60%. [10]

Section B (Answer any EIGHT questions) $[8 \times 5 = 40]$

4. What is an algorithm? How is it different from a flowchart? [5]
5. Explain type conversion and type casting with suitable examples. [5]
6. Explain increment and decrement operators with examples. [5]
7. Write a program that computes sum of digits of an integer. [5]
8. Define function and list advantages of using functions in C programs. [5]
9. Using an array of 10 integers, compute sum and average. [5]
10. Define pointers and explain the relationship between pointers and 1D arrays. [5]
11. Read and display content from a file named `input.txt`. [5]
12. Why are graphics functions used? Write a program to draw a circle. [5]